

Gandhar Anirudha Khandagale

Fairfax, VA

[Email](#) | [GitHub](#) | (571) 495-9456 | [LinkedIn](#)

SUMMARY

Software Engineer with expertise in architecting large-scale distributed systems and intelligent search infrastructures. Proven track record at AWS designing and deploying scalable backend services using Java, Kotlin, and System Design principles. Deeply skilled in Machine Learning, specializing in search ranking algorithms (LambdaMART, ColBERT) and vector retrieval using Python, TensorFlow, and PyTorch. Proficient in cloud-native development on AWS (ECS, DynamoDB), with a strong focus on system reliability, CI/CD pipelines, and performance optimization

EDUCATION

George Mason University, Masters of Science in Computer Science (GPA - 3.67/4.00)

Fairfax, Virginia

University of Mumbai, Bachelors in Computer Engineering (GPA – 3.8/4.0)

Mumbai, India

EXPERIENCE

Software Development Engineer at Amazon Web Services (AWS)

May 2025, Arlington, US

- **System Design & Implementation:** Designed and implemented a new scalable backend service for the EC2 Metering team—a critical organization driving \$42 billion in annual revenue—using Java, Kotlin, and clouddservices.
- **Cloud Architecture & Observability:** Architected a containerized, fault-tolerant system using Docker images and Amazon ECS for orchestration, with SQS for asynchronous messaging and DynamoDB/S3 for storage. Integrated Grafana and CloudWatch to build comprehensive dashboards for real-time system monitoring and observability.
- **Infrastructure & Pipelines:** Automated infrastructure provisioning via CloudFormation (IaC) and managed production CI/CD pipelines, proactively resolving deployment blockages to ensure rapid, reliable feature delivery.
- **Operational Excellence:** Served as a key on-call engineer, troubleshooting complex production incidents and acting as the technical point of contact for cross-team queries to maintain high availability.

Machine Learning Engineer at Neptune Technologies LLC

September 2024 – November 2024, US, Remote

- Developed and deployed ML models for financial risk assessment using TensorFlow, PyTorch, and cloud platform like AWS, integrating them into production-ready backend services with REST APIs.
- Built scalable ML pipelines with Airflow, Spark, and Beam, ensuring high-throughput processing and system reliability..
- Designed real-time fraud detection systems using Autoencoders, Isolation Forest, and integrated with BigQuery, Dataflow for live analytics.
- Managed deployments via AWS CLI, with unit testing, CI/CD, and system diagnostics for robust ops.
- Integrated FAISS, ColBERT, ScaNN for vector search APIs; built LLM-based search evaluation pipelines for relevance tuning.
- Developed search evaluation pipelines using LLMs and vector search for automated relevance and ranking analysis.
- Applied HMMs, LSTMs, XGBoost, GNNs for predictive analytics, time-series modeling, and anomaly detection.
- Collaborated with cross-functional teams in an agile environment to deliver scalable, production-grade AI features.

Machine Learning Intern at Performigence

June 23 – May 24, Los Angeles, CA

- Developed ML models to detect fraudulent activities and abusive behavior within online communities, improving security measures by 30%, while incorporating Fairness Indicators and SHAP for bias detection and explainability.
- Developed NLP-based search query models using BERT, SBERT, and ColBERT, and fine-tuned LLM prompts for query understanding and ranking evaluation, improving text retrieval accuracy in content moderation.
- Engineered LLM prompts and evaluation workflows to assess the relevance of search results, enhancing AI-driven moderation search systems for better precision and reliability.
- Designed LambdaMART-based ranking models for prioritizing flagged abusive content, optimizing search ranking by 20%, while utilizing FairRank and adversarial debiasing techniques to ensure equitable ranking outcomes.

Data Analyst Intern at International Food Policy Research Institute

September 2023 – Present, Fairfax, VA

- Led climate risk assessments focused on climate-related hazards and their impacts on agriculture in Africa and globally using Python and R for data analysis.
- Conducted deep-dive data analyses to improve operational efficiency utilizing Pandas, NumPy, and SQL for data manipulation and processing.
- Performed qualitative and quantitative analysis of operational functions, workflows, and processes with Excel, Tableau, and Power BI for visualization and reporting.
- Collaborated with data engineers and data scientists to ensure data integrity and quality using Git for version control and AWS S3 for data storage.
- Designed and documented dashboards, reports, and thresholds using Tableau and Power BI to monitor business status and fluctuations effectively.

- Conducted DFT simulations using Gaussian16 software to analyze electronic structure properties of metalloporphyrin.
- Developed and implemented a Python-based Graph Neural Network (GNN) model to predict and optimize molecular properties of porphyrin-based compounds. Optimized existing models to enhance performance and accuracy.
- Worked on neural networks for protein peptides and clustering and alignment of peptides for Mouse, Human, and Alligator.
- Collaborated closely with the Biology department to understand and apply key biological concepts to machine learning models.

- Designed and developed software solutions for defect detection in construction materials using computer vision models (TensorFlow, CNNs) integrated into production pipelines.
- Built backend systems and optimization tools for resource allocation using Python and PyTorch, improving efficiency in materials and equipment usage.
- Developed and maintained scalable data pipelines to process real-time data from IoT sensors and devices, leveraging AWS (EC2, S3) for storage and compute scalability.
- Collaborated with cross-functional teams (engineers, managers) to deploy software solutions on-site, ensuring seamless integration with existing workflows and enhancing overall project efficiency.

- Led a machine learning project using TensorFlow, Keras, and Image Visualization to recognize Face Mask, Face Shield, and Gloves.
- Acquired diversified experience in addressing machine breakdowns and overseeing Human-Machine Interface (HMI) tasks. Participated in design reviews to select the best technologies and solutions.

SKILLS

Software Development: Python, C++, C, SQL, HTML, CSS, Bootstrap, PostgreSQL, Familiar with Java and Kotlin, Object-Oriented Design, System Design.

Machine Learning & AI: TensorFlow, PyTorch, Search Quality Evaluation, Scikit-learn, Hugging Face Transformers, Keras, LLM Prompt Engineering, FAISS, Annoy, ScaNN, BM25, ColBERT, LambdaMART, XGBoost, LightGBM.

Scalability & Deployment: Kubernetes, TensorFlow Extended (TFX), TensorFlow Serving, Apache Beam, ONNX, MLflow.

Big Data & Cloud: Google Cloud (Vertex AI, BigQuery, Dataflow, TFX), AWS (SageMaker, Lambda, S3), Apache Spark, Kubernetes, Docker, CI/CD Pipelines, Terraform.

Patents and Publications

Patent Filed: Methods to Identify New Molecules using Machine Learning (Filed: August 9, 2024).

Publications: Face Recognition for the Visually Impaired (2021, 2022), Intelligent Chatbot for College Enquiry (2021), Digital Medical Records with Blockchain (2021).